

PRANITA KHANAL

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Education

University of Texas at Arlington

Bachelor of Science in Computer Science

Arlington, TX

Aug 2023 - May 2027

Relevant Coursework

- Data Structures and Algorithms
- Object-Oriented Programming
- Operating Systems
- Data Bases
- Secure Programming
- Computer Networks and Organization
- Artificial Intelligence
- Neural Network and Deep Learning

Work Experience

Department of Physics

Student Administrative Support

The University of Texas at Arlington

Jan 2024- May 2026

- Manage weekly technical calibration and setup for 54 stations across 6 physics labs; and troubleshoot electronic components to ensure equipment readiness for students.
- Modify lab manuals, create new experiments, turn it into labs and manage inventory logistics by conducting vendor cost-benefit analyses to optimize the departmental budget.
- Oversee the installation and maintenance of lab computers and digital interfaces; lead the integration of PASCO Capstone and Data Studio software to modernize data acquisition and streamline experiment workflows.
- Utilize AutoCAD to engineer custom mechanical components for Mechanics labs; perform hardware repairs, including precision soldering and circuitry to maintain assets.

Deloitte Australia Data Analytics

Forage Online Simulation

Remote

Jan 2026

- Completed a Deloitte job simulation involving data analysis and forensic technology.
- Created a data dashboard using Tableau.
- Used Excel to classify data and draw business conclusions.

JP Morgan Chase and Co. Software Engineering Virtual Experience

Forage Online Simulation

Remote

Jan 2026

- Integrated Kafka into a Spring Boot microservice to consume and deserialize high-volume transaction messages using a configurable topic and embedded Kafka test framework.
- Implemented transaction validation and persistence logic with Spring Data JPA and an H2 SQL database, including entity modeling and balance updates across relational User records.
- Connected the service to an external REST Incentive API using RestTemplate, processing incentive responses and incorporating them into transactional workflows.
- Developed a REST endpoint for querying user balances, returning JSON responses through a Spring controller while maintaining clean architectural boundaries.
- Verified system behavior using Maven test suites and debugger-driven inspection, ensuring reliability.

Projects

RAG PDF Chatbot

Python, CSS, HTML, JS

Dec 2025

pdf-chatbot-frontend-steel.vercel.app

- Developed full-stack AI chatbot application with FastAPI backend and JavaScript frontend, implementing RAG architecture with Pinecone vector database and Google Gemini 2.5 API.
- Deployed containerized microservices on Hugging Face Spaces and Vercel, establishing CI/CD pipeline through GitHub with automated deployments and environment variable management.
- Engineered semantic search system using Sentence Transformers for local embeddings and optimizing API costs while maintaining response accuracy.
- Designed document processing pipeline handling PDF text extraction, chunking, and vectorization, with storage strategy supporting unlimited concurrent users within free tier limits.

Fire Prediction System

Python

Dec 2025

Fire-Prediction@github

- Implemented Random Forest machine learning algorithm using Python and scikit-learn to predict wildfire occurrence from weather condition data.
- Processed and merged multiple weather datasets using pandas, performing data cleaning, feature selection, and regional comparative analysis.
- Created data visualizations with matplotlib including feature importance charts and weather pattern distributions for stakeholder communication.
- Developed interactive prediction interface allowing users to input weather parameters and receive real-time fire risk assessments.

Full-Stack Grocery Store Management System

SQL, PHP

Sep 2025

Databases@github

- Designed a full grocery store database from scratch, turning complex EER diagrams into a working system with 13 different tables.
- Built a custom web dashboard using PHP and SQL so users can easily search for, add, or update inventory items in real-time
- Wrote advanced SQL queries and created views to automatically track store revenue and vendor performance metrics and tested the system against strict integrity rules.

Maverick Online Entertainment System

Java

Nov 2024

MOES@github

- Developed subscription-based media streaming platform with unlimited and pay-per-view account models, managing student authentication and content delivery.
- Implemented persistent data storage using BufferedReader/Writer with automatic backup creation and file validation via magic cookie verification.
- Designed object-oriented architecture leveraging abstract classes, inheritance, and polymorphism for scalable account management with comprehensive regression testing.

Checkers

Java, CSS, HTML, JS (Group Project)

March 2025

- Implemented three gameplay modes (PvP, PvBot, Bot vs. Bot) with clean state logic and AI transitions.
- Produced project artifacts—requirements, state diagrams, UI visuals—for formal documentation.
- Coordinated data handoff to the database team, enabling seamless frontend-backend integration.

Technical Skills

Languages: Python, Java, C, JavaScript, C++, SQL, PHP, Assembly, R, HTML/CSS

Frameworks & Libraries: Spring Boot, Spring Data JPA, Apache Kafka, FastAPI, React.js, RESTful API

AI/ML Tools: Scikit-learn, Pandas, NumPy, Matplotlib, Sentence Transformer

Data Science & Analytics: Data Analysis, Data Modeling, Data Visualization

DevOps & Tools: Docker, Git/Github, CI/CD Pipeline, Maven, Linux/UNIX, API Integration

Leadership / Extracurricular

Project Instructor

UK SARAID, The Japan Foundation

Kathmandu, Nepal

2018- 2020

- Coordinator for SMART Camp, and served as Assistant Instructor for SMART's international COVID-19 session with cross country research teams.
- Instructed one of the HANDS selected Collaborative Action Challenge(CAC) projects, "SMART 2019" for the Kathmandu district which was completed with around 150 participants.